

Scientific Research

Dirk Riehle, FAU Erlangen

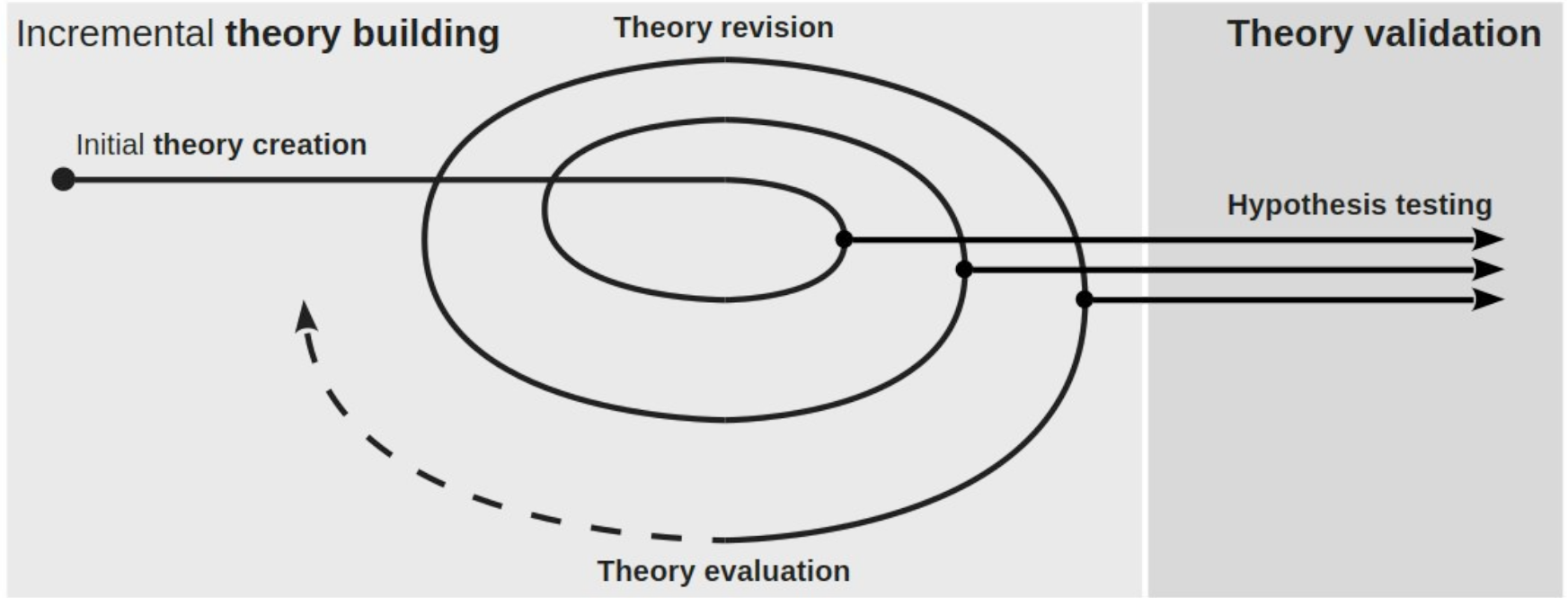
NYT B02

Licensed under [CC BY 4.0 International](https://creativecommons.org/licenses/by/4.0/)

Agenda

1. Research questions
2. Research projects
3. Research methodologies
4. Research methods
5. Research practices
6. Quality criteria

Theory Building and Validation (Recap)



DR

1. Research Questions

Research Question and Hypothesis

A research question is

- The question of interest to be answered by research

A research hypothesis is

- A research question that can only be answered with yes or no, true or false

Research questions should answerable and non-trivial

- The answer can be big (a whole theory) or small (yes/no)

Hilbert's 23 Problems [1]

Problem ↕	Brief explanation	Status ↕	Year Solved ↕
1st	The continuum hypothesis (that is, there is no set whose cardinality is strictly between that of the integers and that of the real numbers)	Proven to be impossible to prove or disprove within the Zermelo–Fraenkel set theory with or without the Axiom of Choice (provided the Zermelo–Fraenkel set theory with or without the Axiom of Choice is consistent, i.e., contains no two theorems such that one is a negation of the other). There is no consensus on whether this is a solution to the problem.	1963
2nd	Prove that the axioms of arithmetic are consistent .	There is no consensus on whether results of Gödel and Gentzen give a solution to the problem as stated by Hilbert. Gödel's second incompleteness theorem , proved in 1931, shows that no proof of its consistency can be carried out within arithmetic itself. Gentzen proved in 1936 that the consistency of arithmetic follows from the well-foundedness of the ordinal ϵ_0 .	1936?
3rd	Given any two polyhedra of equal volume, is it always possible to cut the first into finitely many polyhedral pieces which can be reassembled to yield the second?	Resolved. Result: no, proved using Dehn invariants .	1900
4th	Construct all metrics where lines are geodesics .	Too vague to be stated resolved or not. ^[n 1]	–
5th	Are continuous groups automatically differential groups ?	Resolved by Andrew Gleason , depending on how the original statement is interpreted. If, however, it is understood as an equivalent of the Hilbert–Smith conjecture , it is still unsolved.	1953?

[1] See https://en.wikipedia.org/wiki/Hilbert%27s_problems

Example Research Questions [1]

	People	Practices	Artifacts
Method of Analysis	How to measure communication effectiveness?	How to track bug fixes in software development?	How to measure effort spent on a piece code?
Method of Development	How to distribute roles in an agile development team?	How to perform code reviews?	How does agile development ensure high quality?
Evaluation of Instance	What skills of Linus Torvalds are key for the Linux kernel?	What are some new types of code refactorings?	How patchy is the Apache web server?
Generalization or Characterization	What does it take to be a good programmer?	What's common to all open source processes?	How does the the open source commit distribution look like?
Feasibility or Exploration	Do energy drinks help programmers write code faster?	Is "Test First" a viable principle?	What could a power plant security hole do to safety?

[1] Derived from [Shaw \(2003\)](#): Writing good software engineering research papers.

Related Work

Related work is

- (All) existing scientific work related to your research question

Related means

- Someone else worked on the same or similar research question
- The work takes an alternative or contradicting starting point
- Your proposed work requires and builds on this (related) work

Searching for and identifying

- **Related work is likely to change your research question**

A literature review is a common form of searching for related work

Research Result

A **scientific research result** is

- Any piece of information resulting from scientific research
- Where scientific research is work following proper methods

The **answer** to a research question is

- A scientific result that answers a corresponding research question

A scientific result should be

- verifiable,
- reproducible, and
- independent of the researcher

Paper

A **research paper** is a (science-idiomatic) term for

- A research article

It may or may not yet have passed peer review

Contribution

A **contribution** is a (science-idiomatic) term for

- A publishable / published research result

To be found in the contributions section of a research paper

2. Research Projects

Research Projects

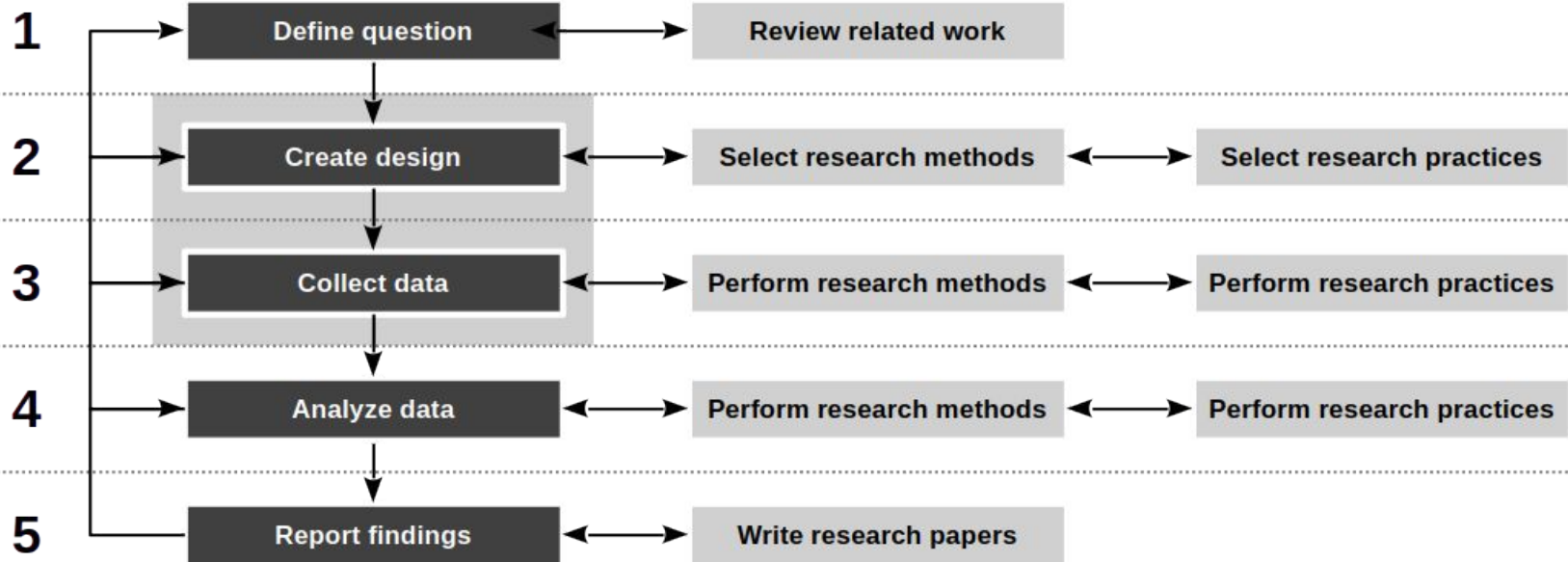
A research project

- Is a project that contributes to scientific progress
- Typically answers a (large or small) question
- As a project has a start and an end date

Research projects can be broken down into subprojects

- A principal investigator may define a whole project
- Graduate researchers may work on subprojects
- Thesis students may work on one question

Generic (Idealized) Research Process



DR

Research Designs

A research design is

- A structured plan to answer a research question
- Created by applying a research method or methodology

A research protocol is

- The documentation of a research design
- Created for quality assurance purposes

Research Methodology vs. Design vs. Method vs. Practice

A research methodology is

- A start-to-finish framework for performing theory building

A research method is

- A method for answering a type of research question

A research practice is

- A way of doing something with a defined outcome

Should You Invent Your Own Methods?

(Almost) never invent your own methods

- Rather use established research methods

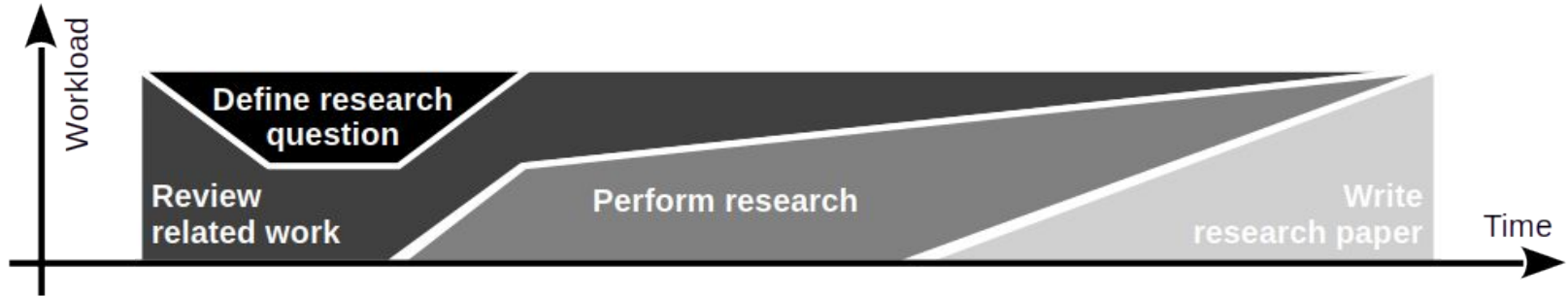
There will always be exceptions

- But think hard if your situation is one

Using an established method instantiates it

- As you near its limits, it becomes less valid

How Time is Spent in Research



DR

(Almost) All Research is Iterative

The scientific process is

- Iterative and incremental in how it builds out theory

Its results presentation should

- Follow a linear logical structure usually corresponding to the research design

Beginners mistake when writing a thesis

- Focusing your process steps, not your results

3. Research Methodologies

Research Methodology

A **research methodology** is a

- Start-to-finish framework for theory building
- Constraint system for research designs
- Pattern for research designs

The Qualitative Survey [1] [2]

1. Define question
2. Create design
3. Collect data
 - a. Build sampling model
 - b. Sample for theory building purposes
 - c. Perform interviews
4. Analyze data
 - a. Analyze transcriptions
 - b. Determine saturation
5. Publish results

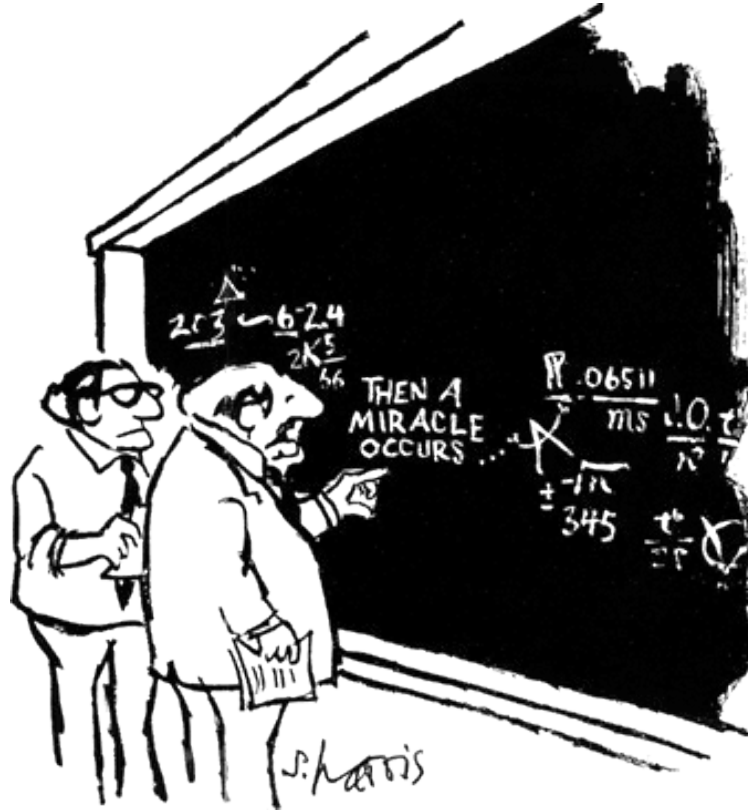
[1] As defined in Jansen (2010)

[2] A.k.a. interview study (when restricted to interviews)

Some (Categories of) Research Methodologies

- Systematic reviews
- Qualitative surveys
- Action research
- Case study research
- Grounded theory
- Ethnographies

“I think you should be more explicit in step 2” [1]



[1] See <http://www.sciencecartoonsplus.com/>

Example Research Design Inner Source Challenges

What type of research methodology is this?

1. Research question

- a. What are challenges of inner source collaboration in software development?

2. Research design

- a. Model units of analysis (companies, employees) of interest
 - b. Identify units of analysis using sampling model
 - c. Contact and interview experts
 - d. Analyze interviews using thematic analysis according to Braun & Clarke (2012)
 - e. Determine saturation and iterate if not satisfied

3. Data collection

4. Data analysis

4. Research Methods

Research Method

A **research method** is

- A prescriptive procedure to answer a research question
- Which collects and analyses one set of data

An **established** research method has been validated

- It has been shown to deliver what it promises
- Methods should have their own quality assurance criteria

Research method vs. **research methodology**

- A methodology provides a framework while a method provides a prescription
- It ain't a methodology if it ain't theory building

Example Research Methods

- Braun & Clarke (2012): Thematic analysis
- Wohlin et al. (2012): Controlled experiments
- Fagerholm & Fritz (2020): Biometric measurements

(A Reenactment of) The Marshmallow Experiment [1]



[1] See <https://www.youtube.com/watch?v=Yo4WF3cSd9Q>

Two Categories of Research Methods

A **qualitative research method** is a research method that

- Collects and analyses qualitative data
- Creates theoretical insights

A **quantitative research method** is a research method that

- Collects and analyses quantitative data
- Lets researchers draw descriptive or statistical conclusions

Qualitative vs. Quantitative Data

Qualitative data is data that

- Usually is not numeric
- Requires interpretation
- Is used in theory building
- Denies statistical generalizations

Quantitative data is data that

- Usually is numeric (various scales)
- Has objective (formal) definition
- Is used for theory validation
- Serves statistical generalization

“Soft” vs. “Hard”

Qualitative research

- Deals with qualitative data
- Feedback takes time
- Feedback is not binary

Quantitative research

- Deals with quantitative data
- Feedback is fast
- Feedback is precise / binary

5. Research Practices

Research Practice

A **research practice** is a

- An established procedure that when employed provides a defined result
- That is known to practitioners of science but has not been codified yet

Research practice vs. **research method**

- A research practice may not have a name or formal definition
- A research practice may be a prototype of a method

Example Research Practices With Name

- Forward snowballing (an article search strategy)
- Polar sampling (a purposive sampling strategy)
- Peer debriefing (a quality assurance practice)

Example Research Practices Without Name

- How to derive a code label
- How to report on a Student's t-test results
- How to codify a theory using best practices handbooks

6. Quality Criteria

Quality Criteria for Research Methods [1]

Intuition	Qualitative research	Quantitative research
Truth value	Credibility	Internal validity
Applicability	Transferability	External validity
Consistency	Dependability	Reliability
Neutrality	Confirmability	Objectivity

[1] See [Guba & Lincoln \(1982\)](#): Naturalistic inquiry.

Truth Value (Category)

Truth value (framed as question)

- To what degree can we establish confidence in the “truth” of findings?

Credibility (naturalism)

- The degree of confidence in the truth of the findings

Internal validity (rationalism)

- The degree of confidence in a cause and effect relationship

Applicability (Category)

Applicability (framed as question)

- To what degree can we determine the applicability of findings in other contexts?

Transferability (naturalism)

- The degree to which findings can be transferred to another context

External validity (rationalism)

- The degree to which findings can be generalized beyond the study

Consistency (Category)

Consistency (framed as question)

- To which degree can the findings be consistently repeated?

Dependability (naturalism)

- The degree to which findings are stable over time

Reliability (rationalism)

- The degree to which findings can be recreated under the same conditions

Neutrality (Category)

Neutrality (framed as question)

- To which degree are the findings independent of the researcher?

Confirmability (naturalism)

- The degree to which the findings can be confirmed by other researchers

Objectivity (rationalism)

- The degree to which the findings can be repeated by other researchers

Summary

1. Research questions
2. Research projects
3. Research methodologies
4. Research methods
5. Research practices
6. Quality criteria

Thank you! Any questions?

dirk.riehle@fau.de – <https://oss.cs.fau.de>

dirk@riehle.org – <https://dirkriehle.com> – [@dirkriehle](#)

Legal Notices

License

- Licensed under the [CC BY 4.0 International](https://creativecommons.org/licenses/by/4.0/) license

Copyright

- © 2012-2025 Dirk Riehle, some rights reserved